

TW11: Cleaning Imported Data

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Profile a dirty retail dataset, justify missing-data decisions, standardise representations, scale features, and validate the final result.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Produce a complete data-quality profile

Scenario

The retail file contains missing values, duplicates, inconsistent categories, mixed dates, and invalid ranges.

INSTRUCTIONS

1. Import `business_sales_dirty.csv` while preserving a raw copy.
2. Report shape, types, present count, missing count and percentage, unique count, and sample values.
3. Count exact duplicates and duplicated order IDs separately.
4. Flag invalid dates, non-positive prices, units outside 1 to 20, and satisfaction outside 1 to 5.

REQUIRED SUBMISSION

- A one-row-per-column profile sorted by missing percentage.
- A separate issue-count report.
- A paragraph identifying the three highest-priority risks.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Design a missing-data strategy

Scenario

Management needs a clean table but wants all treatment assumptions documented.

INSTRUCTIONS

1. Compare missing rates overall and by category or channel.
2. Add indicators for missing region, units, price, and satisfaction.
3. Compare global median and group-median filling for units.
4. Choose retain, fill, source-correct, or drop for each field and justify the choice.
5. Show before-and-after missingness without modifying raw.

REQUIRED SUBMISSION

- A missingness comparison table.
- A cleaned copy with Boolean indicators.
- A decision log containing issue, evidence, action, rationale, and impact.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Repair duplicates, ranges, and labels

Scenario

The analysis needs unique order rows and consistent region and channel categories.

INSTRUCTIONS

1. Remove exact duplicates while recording the number removed.
2. Inspect duplicated order IDs before deciding whether they are exact or conflicting.
3. Trim and title-case region and channel values.
4. Map Telephone to Phone with an explicit dictionary.
5. Mask invalid numerical values before applying any fill strategy.

REQUIRED SUBMISSION

- A cleaned copy and before-after quality table.
- Printed category vocabularies after standardisation.
- Assertions covering uniqueness and numerical ranges.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Parse dates and investigate encoding

Scenario

Dates use more than one format and several entries are invalid; a separate text file uses Latin-1.

INSTRUCTIONS

1. Parse order_date with mixed formats, day-first interpretation, and errors='coerce'.
2. Display every failed date with order ID and original text.
3. Create year, quarter, month, weekday, and weekend features.
4. Decode the Latin-1 notes file and convert its text to UTF-8 bytes.
5. Explain the risk of silently discarding decoding errors.

REQUIRED SUBMISSION

- A date-failure audit.
- Five validated calendar features.
- Correctly decoded text and an encoding-risk explanation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Compare scaling and certify the clean dataset

Scenario

A modelling team needs clean numerical features and proof that the final table meets its contract.

INSTRUCTIONS

1. Create StandardScaler, MinMaxScaler, and RobustScaler versions of units, unit_price, and marketing_spend.
2. Compare min, median, mean, maximum, and standard deviation for each result.
3. Create one-hot channel and region indicators while retaining original labels.
4. Build a final cleaned dataset and write at least eight assertions.
5. Write a recommendation for which scaler to use and why.

REQUIRED SUBMISSION

- A scaling comparison table.
- An encoded but traceable feature table.
- Eight or more passing data-contract assertions and a justified recommendation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____